

SYSTEMS SIMULATION

Systems Analysis & Design

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Basic Concepts
- 2 Cellular Automata
- 3 Digital Twins



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What is a simulation?

- **Simulations** are a type of real-world representation, **but rarely exactly** like the real world.
- Sometimes you need to **test** or **experiment** with **expensive use cases**. Simulations let you **experiment** with different inputs, conditions, and hyperparameter optimizations.
- Also, there are dangerous or hard-to-reach **scenarios** where **simulations** become the best option.
- You should define the **level of detail** well enough to represent the expected behavior, without failing due to either **high complexity** or **oversimplification**.



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Events and Stochastic Processes

- It is typical to work with **event probabilities**, creating **stochastic behaviors**.
- One way to **simulate** many **systems** is to use **event-based models**.
Embrace the chaos and Murphy's Law.



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Case Study: Load Balancer in a Cloud Architecture

- A **load balancer** distributes **incoming requests** across multiple **server instances** to optimize performance and availability.
- **Simulation** helps predict **system behavior** under different traffic loads and failure scenarios.
- Key metrics include **response time**, **throughput**, and **resource utilization**.



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Cellular Automata

- **Cellular Automata** are a **discrete model** consisting of a **grid** of cells, each with a **state**.
- The **state** of each cell is updated based on the **states of its neighbors**.

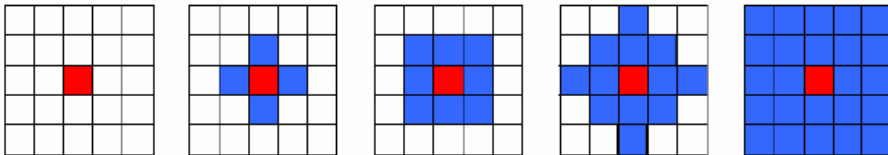
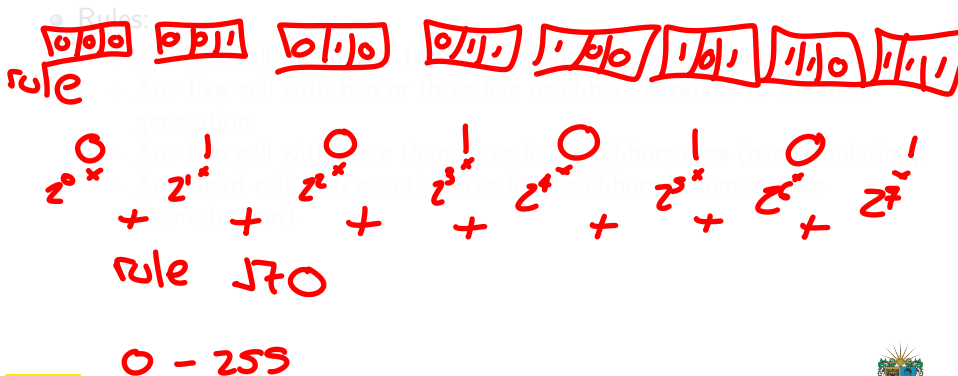


Figure: Cellular Automata Typical Neighborhoods



Game of Life

- **Conway's Game of Life** is a **cellular automaton** devised by the British mathematician **John Horton Conway** in 1970.
- It is a **zero-player game**, meaning that its **evolution** is determined by its **initial state**, requiring no further input.



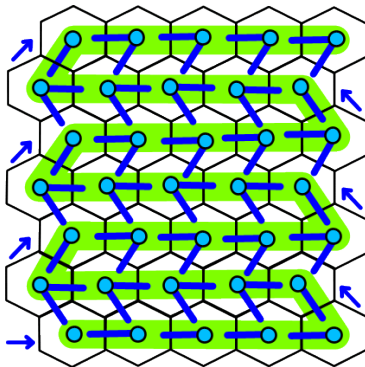
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- It is a **zero-player game**, meaning that its **evolution** is determined by its **initial state**, requiring no further input.
- Rules:
 - Any **live** cell with **fewer than two live** neighbors **dies** (underpopulation).
 - Any **live** cell with **two or three live** neighbors **survives** to the next generation.
 - Any **live** cell with **more than three live** neighbors **dies** (overpopulation).
 - Any **dead** cell with **exactly three live** neighbors becomes **alive** (reproduction).



Honeycomb Cellular Automata

- **Honeycomb Cellular Automata** use a different topology where each cell has six neighbors.
- This representation has different dispersion properties, which can be more interesting for certain applications.



PSP

Graphs



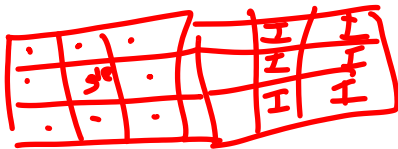
SIR Model

→ COVID-19

- The **SIR Model** is a **compartmental model** used to represent the **transmission of contagious diseases**.
- The **model** divides the population into three compartments: **S** for **susceptible individuals**, **I** for **infected individuals**, and **R** for **recovered individuals**.
- The model is defined by the following **differential equations** where β is the **transmission rate** and γ is the **recovery rate**:

$$\begin{aligned} \bullet \frac{dS}{dt} &= -\beta \cdot S \cdot I + \gamma I \\ \bullet \frac{dI}{dt} &= \beta \cdot S \cdot I - \gamma \cdot I \\ \bullet \frac{dR}{dt} &= \gamma \cdot I \end{aligned}$$

999
↓
I



Chaotic Systems

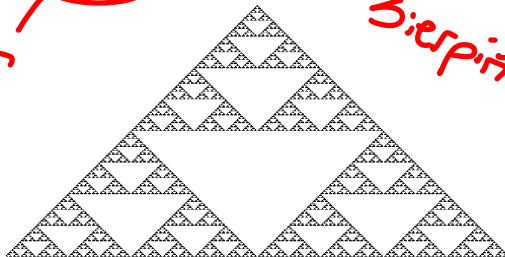
□ 2D ▣ 3D

- **Chaotic Systems** are a class of dynamical systems that exhibit sensitive dependence on initial conditions.
- This means that the future behavior of the system is highly sensitive to the initial conditions.
- The **Lorenz System** is a well-known example of a chaotic system.
- Using **cellular automata** to simulate chaotic systems is common practice. Many fractals can be created using chaotic rules.

attractor

Dimension
2.1
1.7
2.8

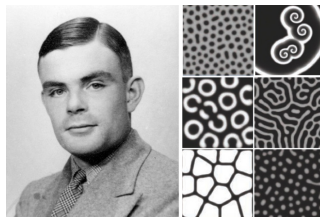
Sierpinski



Turing Morphogenesis

20-30-40 | 45 → Christopher
mind (II) - biology

- **Turing Morphogenesis** is a theory of biological development that explains how spatial patterns form in living organisms.
- The theory is based on the idea that chemical signals can interact to create spatial patterns in living organisms.
- The [reaction-diffusion] model is commonly used to simulate **Turing morphogenesis**.
→ substances
- The model is defined by a set of reaction and diffusion equations that describe how the chemical signals interact.



Outline

1 Basic Concepts

$x+dx$
 $y+dy$

2 Cellular Automata

3 Digital Twins

$$dx = [-1, 0, 1]$$

$$dy = [-1, 0, 1]$$

No
 $dx=0$
 $dy=0$

-1, -1	0, -1	1, -1
-1, 0	•	1, 0
-1, 1	0, 1	1, 1



What are Digital Twins?

- A **Digital Twin** is a virtual model that accurately reflects a physical object or system.
 Simulation
- **Digital twins** enable real-time monitoring, simulation, and analysis of the physical counterpart.
 2.0
- They are widely used in manufacturing, smart cities, healthcare, and energy systems.
 Simulation + Compute
- By integrating sensor data, **digital twins** help in predictive maintenance and process optimization.
 power

*IOT → Temperature
Cars
TVs
Biomedical Signals*

*frequency
30 seconds
5 minutes*

*CPU's
GPU's
A.I.*



Applications of Digital Twins

- **Manufacturing:** Monitor equipment performance and predict failures.
- **Urban Planning:** Model city infrastructure to optimize traffic flow and resource allocation.
- **Healthcare:** Simulate patient-specific treatments and optimize medical procedures.
- **Energy Systems:** Analyze grid performance and integrate renewable energy sources.



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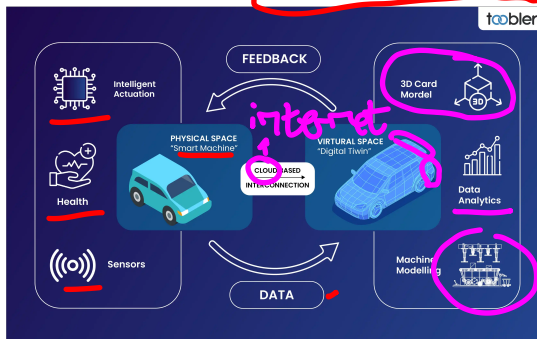


Benefits of Digital Twins

- Enhanced **decision-making** through **real-time data analysis**.
- Improved efficiency and reduced operational costs.
- Better understanding** of **system behavior** and **performance characteristics**.
- Enhanced collaboration** between **teams and stakeholders**.

data science

exports



Challenges of Digital Twins

- Data integration and management.

- Ensuring data security and privacy.

- Maintaining the accuracy of the digital twin.

- Managing the complexity of the physical system.

- Ensuring interoperability between different systems and platforms.

- Scalability to accommodate large-scale systems.

- Cost of implementation and maintenance.

- User training and adoption.

- Regulatory compliance and standardization.

- Ethical considerations in data usage and analysis.

- Environmental impact of digital twin technologies.

Volume
Velocity
Variety

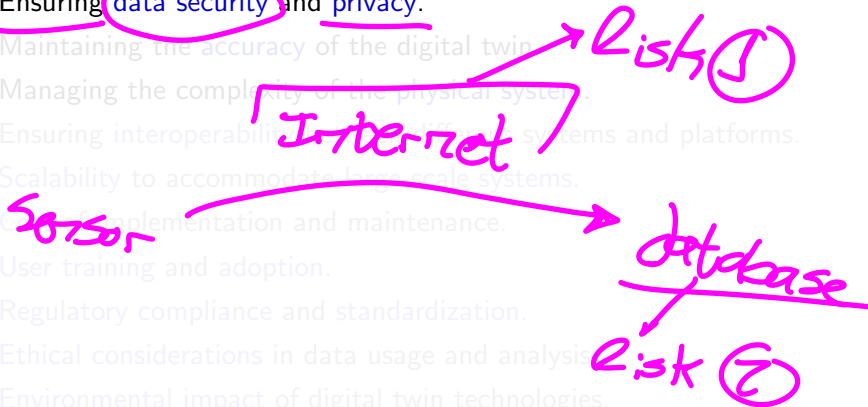
→ 3V

Big Data



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not feasible



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architectures → \$ expensive



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cloud computing + servers



Challenges of Digital Twins

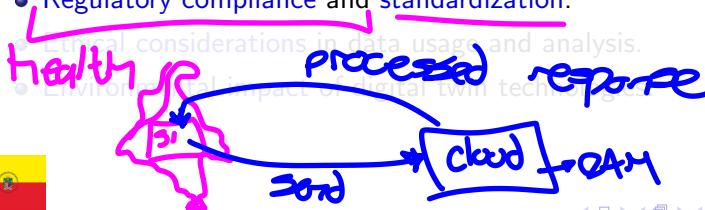
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users



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Cambridge Analytics



Challenges of Digital Twins

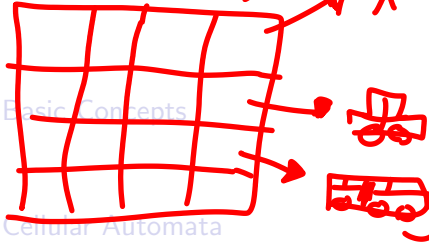
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DataCenters → 2% energy → CO₂



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(RIS)



Objects

class Person:
 n_cont = 0
 rate_vesibore = 0.5
 days_last_i = 5
 rate_contagious = 0.3

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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

